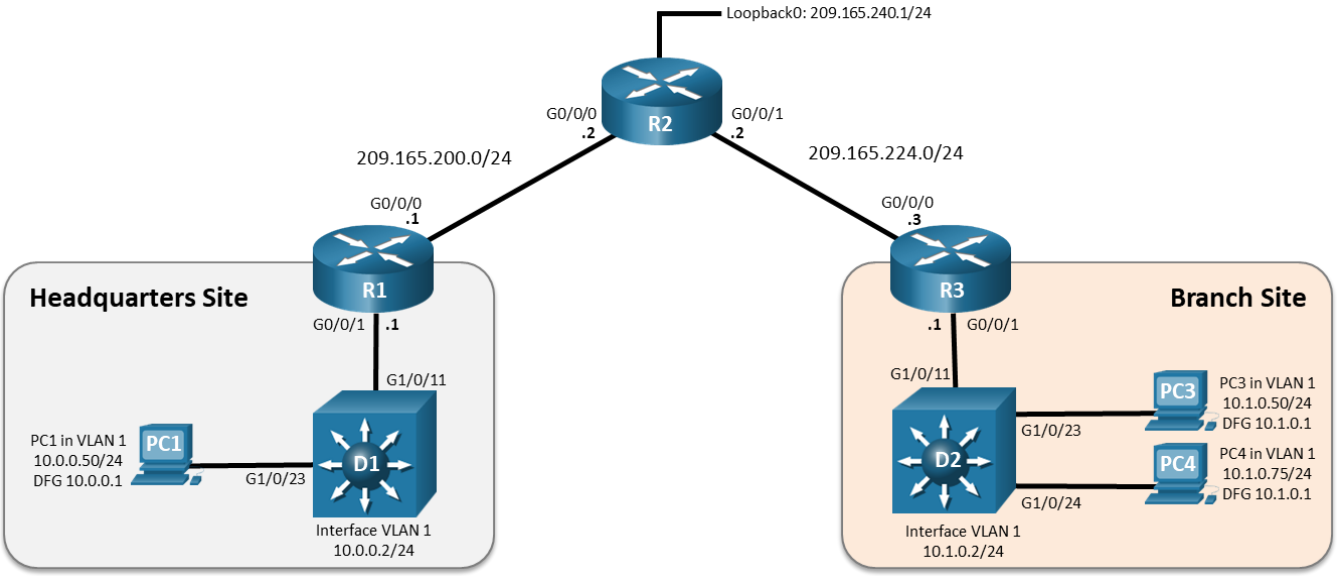
Lab - Implement NAT

Topology



Addressing Table

Device	Interface	IP Address	Default Gateway
R1	G0/0/0	209.165.200.1/24	N/A
	G0/0/1	10.0.0.1/24	
R2	G0/0/0	209.165.200.2/24	N/A
	G0/0/1	209.165.224.2/24	
	Loopback0	209.165.240.1/24	
R3	G0/0/0	209.165.224.3/24	N/A
	G0/0/1	10.1.0.1/24	
D1	VLAN 1	10.0.0.2/24	N/A
D2	VLAN 1	10.2.0.2/24	N/A
PC1	NIC	10.0.0.50/24	10.0.0.1
PC3	NIC	10.1.0.50/24	10.1.0.1
PC4	NIC	10.1.0.75/24	10.1.0.1

Objectives

Part 1: Build the Network and Configure Basic Device Settings

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Part 2: Configure and Verify Static Inside NAT

Part 3: Configure and Verify Dynamic NAT

Part 4: Configure and Verify PAT

Background / Scenario

The HQ and Branch sites must be configured to support NAT. Specifically, the HQ and Branch routers will be configured to provide inside LAN users with outside public addresses using NAT. The HQ router will also provide static NAT to access the email server from the outside network.

Note: This lab is an exercise in configuring and verifying various methods of NAT and does not reflect networking best practices.

Note: The routers and switches used with CCNP hands-on labs are Cisco 8200L and Cisco 9300L, both with Cisco IOS XE Release 17.12.05 (universalk9 image). Other routers and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Ensure that the routers and switches have been erased and have no startup configurations. If you are unsure contact your instructor.

Required Resources

- 3 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)
- 2 Switches (Cisco 9300L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- 3 PCs (Choice of operating system with a terminal emulation program installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet and serial cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram and cable as necessary.

Step 2: Configure basic settings for each device.

a. Console into each device, enter global configuration mode, and apply the basic settings. A command list for each device is provided below for the initial configurations.

Router R1

```
hostname R1
no ip domain lookup
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
```

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```
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
banner motd # This is R1, Implement NAT Lab #
interface g0/0/0
 ip address 209.165.200.1 255.255.255.0
 no shut
 exit
interface g0/0/1
 ip address 10.0.0.1 255.255.255.0
 no shut
 exit
 ip route 0.0.0.0 0.0.0.0 g0/0/0 209.165.200.2
```

Router R2

```
hostname R2
no ip domain lookup
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
banner motd # This is R2, Implement NAT Lab #
interface g0/0/0
ip address 209.165.200.2 255.255.255.0
no shut
exit
interface g0/0/1
ip address 209.165.224.2 255.255.255.0
no shut
exit
interface loopback 0
ip address 209.165.240.1 255.255.255.0
no shut
exit
ip route 10.0.0.0 255.255.255.0 g0/0/0 209.165.200.1
ip route 10.1.0.0 255.255.255.0 g0/0/1 209.165.224.3
```

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Router R3

```
hostname R3
no ip domain lookup
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
banner motd # This is R3, Implement NAT Lab #
interface g0/0/0
 ip address 209.165.224.3 255.255.255.0
 no shut
 exit
interface g0/0/1
 ip address 10.1.0.1 255.255.255.0
 no shut
 exit
ip route 0.0.0.0 0.0.0.0 g0/0/0 209.165.224.2
```

Switch D1

```
hostname D1
no ip domain lookup
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
banner motd # This is D1, Implement NAT Lab #
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface range g1/0/11, g1/0/23
 no shutdown
```

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```
 exit
interface g1/0/23
 switchport mode access
 spanning-tree portfast
 exit
interface vlan 1
 ip address 10.0.0.2 255.255.255.0
 no shutdown
ip default-gateway 10.0.0.1
username admin secret cisco123
line vty 0 15
 login local
```

Switch D2

```
hostname D2
no ip domain lookup
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
banner motd # This is D2, Implement NAT Lab #
interface range g1/0/1-24, g1/1/1-4, g0/0
 shutdown
 exit
interface range g1/0/11, g1/0/23-24
 no shutdown
 exit
interface range g1/0/23-24
 switchport mode access
 spanning-tree portfast
 exit
interface vlan 1
 ip address 10.1.0.2 255.255.255.0
 no shutdown
ip default-gateway 10.1.0.1
```

b. Set the clock on each device to UTC time.

c. Save the running configuration to startup-config.

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Step 3: Verify reachability.

PC1 (10.0.0.50) should be able to ping PC3 (10.1.0.50) and PC4 (10.1.0.75). PC4 should be able to ping switch D1 (10.0.0.2). If not, troubleshoot as necessary.

Part 2: Configure and Verify Static Inside NAT

In Part 2, you will configure and verify Static Inside NAT. The idea behind Static Inside NAT is to make an inside local address reachable via an outside global address. For this lab, we will make interface VLAN 1 on switch D1 appear as 209.165.200.99 on the outside network.

Step 1: On R1, configure Static Inside NAT.

Note: The following steps a and b do not have to be carried out in the order listed.

a. Configure R1 to translate the address on D1 Interface VLAN 1 to 209.165.200.99.

```
R1(config)# ip nat inside source static 10.0.0.2 209.165.200.99
```

b. On R1, specify the inside and outside interfaces for NAT purposes.

```
R1(config)# interface g0/0/0
R1(config-if)# ip nat outside
R1(config-if)# exit
R1(config)# interface g0/0/1
R1(config-if)# ip nat inside
R1(config-if)# exit
```

Step 2: Verify the NAT process is occurring on R1.

a. On R1, issue the command **show ip nat translations**. In the output, you will see the static translation information.

```
R1# show ip nat translations
Pro  Inside global         Inside local          Outside local         Outside global
---  209.165.200.99        10.0.0.2              ---                   ---
Total number of translations: 1
```

b. From the console of R2, send 10,000 pings to the destination address 209.165.200.99 using the command **ping 209.165.200.99 repeat 10000**. The pings should be successful.

c. On R1, issue the command **show ip nat translations**. In the output, you will see the static translation as well as the translation used for the ping.

```
R1# show ip nat translations
Pro  Inside global         Inside local          Outside local         Outside global
---  209.165.200.99        10.0.0.2              ---                   ---
icmp 209.165.200.99:0      10.0.0.2:0            209.165.200.2:0       209.165.200.2:0
Total number of translations: 2
```

d. From the console of R2, stop the ping if it is still running and then establish an SSH session to 209.165.200.99. You should be able to connect and login to D1. Use the username **admin** and **cisco123** as the password when prompted.

e. While logged in to D1, issue the command **show tcp brief**. In the output, you will see the addresses involved in the communication from D1's perspective.

```
D1# show tcp brief
TCB        Local Address           Foreign Address        (state)
054D9734   10.0.0.2.23             209.165.200.2.63955    ESTAB
```

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f. On R1, issue the command **show ip nat translations**. In the output, you will see the static translation as well as the translation used for the SSH session.

```
R1# show ip nat translations
Pro  Inside global         Inside local   Outside local         Outside global
---  209.165.200.99        10.0.0.2       ---                   ---
tcp  209.165.200.99:22     10.0.0.2:22    209.165.200.2:63955   209.165.200.2:63955
Total number of translations: 2
```

g. On R2, enter **exit** to disconnect the SSH session to D1 in preparation for the next part of the lab.

Part 3: Configure and Verify Dynamic NAT

In Part 3, you will configure, examine, and verify dynamic NAT. Dynamic NAT uses a pool of available outside addresses to dynamically translate inside addresses on a one-to-one basis. The drawback to dynamic NAT is that if you do not have a pool of addresses that is at least the same size as the number of addresses that need to be translated, there will be inside hosts that cannot send traffic outside until a translation times out or is manually cleared.

Step 1: On R3, configure Dynamic NAT.

Note: The following steps a, b, c, and d do not have to be carried out in the order listed.

a. On R3, create a standard access list that identifies the source addresses of traffic to be translated. Traffic with source addresses matching this access list are referred to as “interesting” and will be processed through the configured NAT rule. For this lab, specify the entire 10.1.0.0/24 network as interesting.

```
R3(config)# access-list 33 permit 10.1.0.0 0.0.0.255
```

b. Next, create a pool of addresses to use for the interesting traffic translations. You should have authority to use these addresses, and outside networks should route traffic destined to these addresses to you. For this lab, we will use two addresses from the subnet connecting R3 and R2.

```
R3(config)# ip nat pool DYNAMIC-NAT 209.165.224.5 209.165.224.6 prefix-length 24
```

c. Associate the interesting addresses with the NAT pool.

```
R3(config)# ip nat inside source list 33 pool DYNAMIC-NAT
```

d. Establish the inside and outside interfaces on R3.

```
R3(config)# interface g0/0/0
R3(config-if)# ip nat outside
R3(config-if)# exit
R3(config)# interface g0/0/1
R3(config-if)# ip nat inside
R3(config-if)# exit
```

Step 2: Verify the NAT process is occurring on R3.

a. On R3, issue the command **show ip nat translations**. The output will indicate that there are no translations.

b. On R3, issue the command **show ip nat pool name DYNAMIC-NAT**. The output will give you details about the pool that is available for translation.

```
R3# show ip nat pool name DYNAMIC-NAT

NAT Pool Statistics
```

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```
Pool name DYNAMIC-NAT, id 1
                            Assigned            Available
  Addresses                        0                    2
  UDP Low Ports                    0                 1024
  TCP Low Ports                    0                 1024
  UDP High Ports                   0               129024
  TCP High Ports                   0               129024

(Low ports are less than 1024. High ports are greater than or equal to 1024.)
```

c. From the command prompt on PC3, start a continuous ping to the destination address 209.165.224.2. The pings should be successful.

d. On R3, issue the command **show ip nat translations**. In the output you will see the pooled translation as well as the translation used for the ping.

```
R3# show ip nat translations
Pro  Inside global         Inside local          Outside local         Outside global
---  209.165.224.5         10.1.0.50             ---                   ---
icmp 209.165.224.5:1       10.1.0.50:1           209.165.224.2:1       209.165.224.2:1
Total number of translations: 2
```

e. From the command prompt on PC4, start a continuous ping to the destination address 209.165.224.2. The pings should be successful.

f. On R3, issue the command **show ip nat translations**. In the output you will see the dynamic translations as well as the translation used for the ping.

```
R3# show ip nat translations
Pro  Inside global         Inside local      Outside local         Outside global
---  209.165.224.6         10.1.0.75         ---                   ---
---  209.165.224.5         10.1.0.50         ---                   ---
icmp 209.165.224.6:6740    10.1.0.75:6740    209.165.224.2:6740    209.165.224.2:6740
icmp 209.165.224.5:1       10.1.0.50:1       209.165.224.2:1       209.165.224.2:1
Total number of translations: 4
```

g. From the console of D2, send 10,000 pings to the destination address 209.165.224.2 using the command **ping 209.165.224.2 repeat 10000**. The pings should fail.

h. On R3, you should see the following syslog message being repeated:

```
Sep 29 21:53:17.285: %NAT-6-ADDR_ALLOC_FAILURE: Address allocation failed; pool 1 may
be exhausted [2] port range: NA, non-PATable: NO, for ALG: NO, input intf:
GigabitEthernet0/0/1, mapping-id: 1, created by pkt: src_ip 10.1.0.2 dst_ip
209.165.224.2 src_port 0 dst_port 0 proto 1QFP:0.0 Thread:000 TS:00000003573907612846
```

We purposely created a pool of addresses that was too small to demonstrate one of the shortcomings of dynamic NAT. Verify the pool is exhausted by examining the output of the command **show ip nat pool name DYNAMIC-NAT**.

```
R3# show ip nat pool name DYNAMIC-NAT

NAT Pool Statistics

Pool name DYNAMIC-NAT, id 1
                            Assigned            Available
  Addresses                        2                    0
  UDP Low Ports                    0                 1024
```

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```
  TCP Low Ports                    0                1024
  UDP High Ports                   0              129024
  TCP High Ports                   0              129024

(Low ports are less than 1024. High ports are greater than or equal to 1024.)
```

i. Stop the pings on D2, PC3, and PC4. (**Note**: To stop the ping, press Ctr+Shift+6.)

j. In preparation for the next part of the lab, remove the pool and mapping commands. Leave the ACL and the interface specifications.

```
R3# clear ip nat translation *
R3# conf t
R3(config)# no ip nat inside source list 33 pool DYNAMIC-NAT
R3(config)# no ip nat pool DYNAMIC-NAT 209.165.224.5 209.165.224.6 prefix-
length 24
```

Part 4: Configure and Verify PAT

In Part 4, you will configure and examine NAT Overload, which is also called Port Address Translation or PAT. PAT overloads on an interface address or on a pool of addresses by associating each traffic session with a unique set of port numbers. The advantage, especially for smaller networks, is that everyone can get out of the network by leveraging a single outside IP address. The drawback to PAT is that you have a limit of about 65,000 to the number of translations possible via a single address. The impact of this limit is felt in large networks, where you must have multiple outside IP addresses to overload onto to ensure everyone can communicate.

As a reminder, R3 still has interfaces setup as outside and inside, and still has access list 33 specifying the LAN network address. You will leverage those existing configurations to complete this part of the lab.

For this lab, R3 will overload on the interface specified as outside, which is g0/0/0.

a. Configure the NAT statement directing the router to translate addresses matching access list 33 to the outside interface IP address, keying on port numbers.

```
R3(config)# ip nat inside source list 33 interface g0/0/0 overload
```

b. On R3, issue the command **show ip nat translations**. The output will indicate that there are no translations.

c. From the command prompt on PC3, start a continuous ping to the destination address 209.165.224.2. The pings should be successful.

d. On R3, issue the command **show ip nat translations**. In the output you will see the PAT translation used for the ping.

```
R3# show ip nat translations
Pro  Inside global         Inside local          Outside local         Outside global
icmp 209.165.224.3:1       10.1.0.50:1           209.165.224.2:1       209.165.224.2:1
Total number of translations: 1
```

e. From the command prompt on PC4, start a continuous ping to the destination address 209.165.224.2. From the console of D2, send 100,000 pings to the same address. The pings should be successful.

f. On R3, issue the command **show ip nat translations**. In the output you will see the PAT translations as well as the translation used for the ping from D2 and PC4.

```
R3# show ip nat translations
Pro  Inside global         Inside local       Outside local         Outside global
icmp 209.165.224.3:6791    10.1.0.75:6791     209.165.224.2:6791    209.165.224.2:6791
icmp 209.165.224.3:1       10.1.0.50:1        209.165.224.2:1       209.165.224.2:1
```

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```
icmp 209.165.224.3:6784    10.1.0.2:1          209.165.224.2:1        209.165.224.2:6784
Total number of translations: 3
```

e. Stop the continuous pings on D2, PC3, and PC4.

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